H Roll No. .....

TCE–801

B. TECH. (CE) (EIGHTH SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

CONSTRUCTION PLANNING AND SCHEDULING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Answer any *one* sub question among (a) and (b) in each main question.

(iii) Each sub-question carries 10 marks.

1. (a) Explain 'Head event,' 'Tail event,' 'Dual role event,' 'Successor event' and 'Predecessor event.' (CO1)

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OR

(b) Explain why planning is necessary. Describe various steps for planning a project. (CO1)

2. (a) Differentiate between PERT and CPM Network. (CO1)

OR

(b) Discuss the various phases of project management. (CO1)

3. (a) An activity in CPM network has a duration of 4 days. The free float for the activity is 10 days and the total float is also 10 days. Find the maximum delay that can be allowed for the activity from occurrence of the preceding event. (CO1)

OR

(b) What is the difference between 'free float' and 'total float' ? Explain the significance of each term. (CO1)

4. (a) Determine the output of a bulldozer for the following situations : (CO2)

Material handed sandy loam top soil having swell = 25%

Haul distance = 30 meter

Rated mould board capacity = 3 cumec loose volume

Actual operation time per hours = 45 min.

Forwarded speed = 2.4 km per hours

Reverse speed = 6 km per hour.

OR

(b) Discuss the cost components you will consider in deciding the cost of owing and operation. (CO2)

5. (a) Define the following terms : (CO2, CO1)

(i) Pilling Hammer

(ii) Pump

(iii) Compressor

(iv) Dumper

(v) Bulldozer

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OR

(b) Define the following terms : (CO2, CO1)

(i) Crash Cost

(ii) Crash Duration

(iii) Indirect Cost

(iv) Direct Cost

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